Minikube installation

Minikube is a tool that is used to run Kubernetes locally. It runs a single-node Kubernetes cluster so that hands-on work can be done on Kubernetes for daily software development.

We will use minikube and verify the usage of Lens. Let's install minikube on a Windows based system first. You can also install it on other operating systems, virtual machines or laptops.

We need either a Docker container or a virtual machine environment. The prerequisites are:

- Two or more CPUs
- 2GB of RAM
- 20GB of free disk space
- Internet connectivity
- Container or virtual machine manager such as Docker or VirtualBox

From a terminal or command prompt, execute the minikube *start* command.

```
minikube start --driver=virtualbox
* minikube v1.12.3 on Microsoft Windows 10 Home Single
Language 10.0.19044 Build 19044
* Using the virtualbox driver based on existing profile
* minikube 1.26.0 is available! Download it: https://github.
com/kubernetes/minikube/releases/tag/v1.26.0
* To disable this notice, run: 'minikube config set
WantUpdateNotification false'

* Starting control plane node minikube in cluster minikube
* virtualbox "minikube" VM is missing, will recreate.
* Creating virtualbox VM (CPUs=2, Memory=3000MB,
Disk=20000MB) ...
! This VM is having trouble accessing https://k8s.gcr.io
* To pull new external images, you may need to configure
a proxy: https://minikube.sigs.k8s.io/docs/reference/
networking/proxy/
* Preparing Kubernetes v1.18.3 on Docker 19.03.12 ...
* Verifying Kubernetes components...
* Enabled addons: default-storageclass, storage-provisioner
* Done! kubectl is now configured to use "minikube"
```

Go to your virtual box and verify the newly created minikube virtual machine (Figure 1).

Now verify the existing status of minikube using the *minikube status* command.

```
C:\>minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Next, use the *kubectl cluster-info* command to get details about kubeDNS.

```
kubectl cluster-info
Kubernetes master is running at https://192.168.99.103:8443
KubeDNS is running at https://192.168.99.103:8443/api/v1/
```

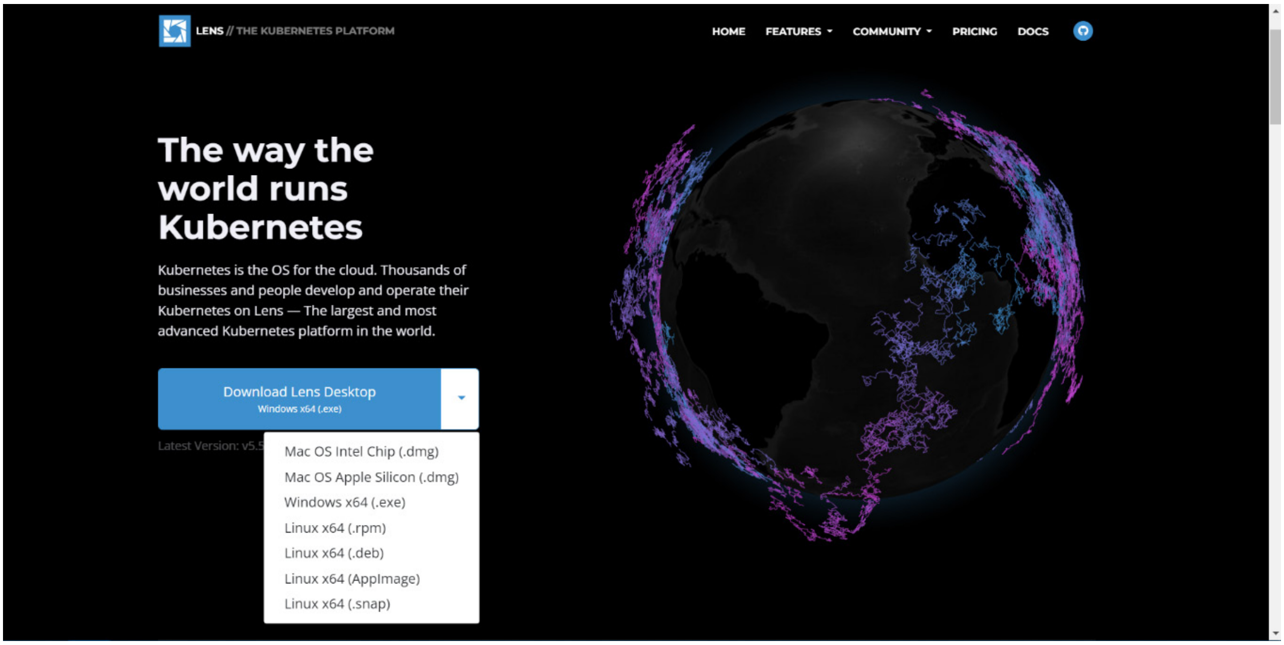

Figure 2: Lens